Research Paper Roadmap for
Cross-Dataset Object Detection in Autonomous Driving

Training on KITTI and External Validation on a Representative Waymo Sample

1. Problem Definition

- **Research Focus:** The goal is to conduct a strong comparative study of object detection models for autonomous-driving perception, using KITTI for training and in-domain validation and a representative Waymo subset for external validation.
- **Key Objective:** The study should evaluate not only detection accuracy, but also speed, computational efficiency, robustness, and cross-dataset generalization.
- **Contextual Challenge:** How can object detection models trained on a smaller autonomous-driving benchmark such as KITTI maintain reliable performance when exposed to a larger and more diverse dataset such as Waymo?
- **Publication Target:** The paper should be framed for IEEE Transactions on Intelligent Vehicles (T-IV), emphasizing automated-vehicle perception, real-time constraints, safety-relevant failure modes, and deployment readiness.

2. Core Research Questions

- How do representative detector families perform on in-domain KITTI validation data?
- How well do KITTI-trained detectors generalize to a representative Waymo front-camera subset without retraining?
- Which detector provides the strongest accuracy-speed-efficiency trade-off for autonomous-driving perception?
- Which object classes suffer the largest performance degradation under domain shift: vehicles, pedestrians, or cyclists?
- How do the detectors behave under safety-critical conditions such as small objects, occlusion, crowded scenes, and false negatives?
- Can a generalization-oriented evaluation protocol produce more deployment-relevant insight than a single-dataset benchmark?

3. Literature Review

- **Autonomous-Driving Object Detection:** Review 2D perception studies on KITTI, Waymo, nuScenes, and other driving datasets, with attention to vehicle, pedestrian, and cyclist detection.
- **One-Stage Real-Time Detection:** Study YOLO-family detectors and their role in low-latency perception pipelines.
- **Two-Stage Detection:** Review Faster R-CNN and region-proposal-based detectors as strong accuracy-oriented baselines.
- **Dense Detection and Focal Loss:** Review RetinaNet and the class-imbalance problem in dense object detection.
- **Transformer-Based Detection:** Review DETR-style models and RT-DETR as an end-to-end real-time transformer detector.
- **Cross-Dataset Generalization:** Analyze work on dataset bias, domain shift, and external validation in autonomous-driving perception.
- **Deployment and Safety-Oriented Evaluation:** Investigate latency, FPS, false negatives, small-object detection, and class-wise robustness as practical AV metrics.

4. Methodology

1. Experimental Architecture:

- **Training Dataset:** Use KITTI as the main training and in-domain validation benchmark.
- **External Validation Dataset:** Use a representative sample from Waymo, preferably front-camera images, to test generalization without retraining.
- **Unified Task:** Use 2D object detection for fair comparison across YOLO, Faster R-CNN, RetinaNet, and RT-DETR.

2. Object Detection Models:

- **YOLO:** Real-time one-stage detector for speed and deployment-oriented analysis.
- **Faster R-CNN:** Two-stage detector for region-based accuracy-focused comparison.
- **RetinaNet:** Dense one-stage detector using focal loss, useful for class imbalance analysis.
- **RT-DETR:** Real-time transformer detector for modern attention-based object detection.

3. Dataset Harmonization:

- **Class Mapping:** Map classes into Vehicle, Pedestrian, and Cyclist. KITTI labels such as Car, Van, and Truck may be mapped to Vehicle if the study design requires it.
- **Annotation Format:** Convert KITTI and Waymo labels into COCO format for shared evaluation and model-specific formats where required.
- **Camera Alignment:** Use Waymo front-camera images first to reduce perspective mismatch with KITTI.

4. Evaluation Strategy:

- **Accuracy Metrics:** mAP@0.5, mAP@0.5:0.95, precision, recall, F1-score, and per-class AP.
- **Efficiency Metrics:** FPS, inference time, parameters, FLOPs, and model size.
- **Generalization Metrics:** KITTI-to-Waymo performance drop and Generalization Ratio = mAP_Waymo / mAP_KITTI.
- **Safety-Oriented Metrics:** False negatives, small-object detection, occlusion performance, and class-wise degradation.

5. Datasets and Tools:

- **Datasets:** KITTI object detection benchmark for training and validation; Waymo Open Dataset subset for external validation.
- **Frameworks:** Ultralytics or official implementation for YOLO; Detectron2 for Faster R-CNN and RetinaNet; official or supported pipeline for RT-DETR.
- **Implementation:** PyTorch-based training and evaluation with version-controlled scripts and reproducible configuration files.

5. Final Four-Model Setup

Detector	Family	Role in the Paper
YOLO	Real-time one-stage CNN detector	Speed/deployment baseline for AV perception
Faster R-CNN	Two-stage CNN detector	Accuracy-focused classical baseline
RetinaNet	Dense one-stage detector	Focal-loss/class-imbalance baseline
RT-DETR	Real-time transformer detector	Modern transformer-based baseline

8-Milestone Roadmap for Cross-Dataset Object Detection in Autonomous Driving

IEEE Transactions on Intelligent Vehicles (T-IV) Target Journal Standards (Q1 Venue)

Milestone 1: Literature Review & Research Question Definition

- **Objective:**
  - **Conduct a comprehensive literature review focusing on:**
  - **Autonomous-Driving Object Detection:** Review KITTI, Waymo, and benchmark studies in vehicle perception.
  - **Detector Families:** Study YOLO, Faster R-CNN, RetinaNet, and RT-DETR as representative architectures.
  - **Domain Shift and Generalization:** Review studies that evaluate how models behave across datasets.
  - **Safety-Oriented Evaluation:** Study metrics related to false negatives, small objects, occlusion, latency, and vulnerable road users.
  - Define the research problem, novel contributions, hypotheses, and T-IV-aligned framing.
- **Deliverables:**
  - Literature review section draft of 2-3 pages.
  - Research questions aligned with autonomous-driving perception and cross-dataset evaluation.
  - Clear contribution list emphasizing accuracy, efficiency, robustness, and generalization.
  - Initial title and abstract draft.

Milestone 2: Dataset Design & Class Harmonization

- **Objective:**
  - **Prepare KITTI as the main benchmark:** Use KITTI for training and in-domain validation.
  - **Construct the Waymo subset:** Use a representative front-camera sample for external validation without retraining.

- **Define class mapping:** Map labels into Vehicle, Pedestrian, and Cyclist to support fair evaluation.
- **Design sampling criteria:** Balance Waymo samples across class distribution, object size, scene density, occlusion, and lighting where available.
- Create a transparent dataset protocol so reviewers can understand exactly how the external validation set was built.

- **Deliverables:**

  - KITTI preprocessing and train/validation split.
  - Waymo subset selection protocol.
  - Class mapping table between KITTI and Waymo.
  - Dataset statistics tables and sample annotation visualizations.
  - Dataset description subsection for the paper.

Milestone 3: Preprocessing Pipeline & Annotation Conversion

- **Objective:**

  - **Standardize annotation formats:** Convert KITTI and Waymo labels into COCO format for shared evaluation.
  - **Prepare model-specific formats:** Generate YOLO format where needed while keeping COCO as the common reference format.
  - **Normalize image handling:** Define resizing, padding, augmentation, and image-quality rules consistently across models.
  - **Verify labels visually:** Inspect bounding boxes for vehicles, pedestrians, and cyclists to catch conversion errors early.
  - Build a reproducible data pipeline that can be shared as part of the paper artifacts.

- **Deliverables:**

  - Annotation conversion scripts.
  - Unified dataset configuration files.
  - Preprocessing documentation.
  - Visual verification samples for KITTI and Waymo.
  - Data preprocessing subsection draft.

Milestone 4: Model Implementation & Fair Training Protocol

- **Objective:**
    - **Implement the four selected models:** YOLO, Faster R-CNN, RetinaNet, and RT-DETR.
    - **Use consistent model scales:** Avoid unfair comparisons such as a small YOLO model against a large RT-DETR model.
    - **Define training fairness rules:** Report input resolution, pretraining, epochs, optimizer, batch size, learning rate schedule, and augmentation policy.
    - **Use reproducible settings:** Fix random seeds and use the same hardware setup for all models where possible.
    - Draft the experimental setup in a way that prevents reviewer criticism about unfair benchmarking.
- **Deliverables:**
    - Training configs for all four models.
    - Hardware/software environment table.
    - Hyperparameter table.
    - Fairness checklist.
    - Model implementation subsection draft.

Milestone 5: KITTI Training & In-Domain Evaluation

- **Objective:**
    - **Train all four models on KITTI:** Use the same training/validation split and consistent evaluation setup.
    - **Evaluate in-domain performance:** Report mAP@0.5, mAP@0.5:0.95, precision, recall, F1-score, and per-class AP.
    - **Measure efficiency:** Report FPS, inference time, parameters, FLOPs, and model size.
    - **Perform class-wise analysis:** Compare performance on vehicles, pedestrians, and cyclists separately.
    - Establish a strong baseline before moving to cross-dataset validation.

- **Deliverables:**
  - KITTI results table.
  - Per-class AP table.
  - Precision-recall curves.
  - FPS and inference-time comparison.
  - Qualitative detection examples on KITTI.
  - In-domain results subsection draft.

Milestone 6: Waymo External Validation & Generalization Analysis

- **Objective:**
  - **Evaluate without retraining:** Apply the KITTI-trained models directly to the Waymo representative subset.
  - **Measure domain-shift degradation:** Compare KITTI validation performance against Waymo external validation performance.
  - **Compute Generalization Ratio:** Use Generalization Ratio = mAP_Waymo / mAP_KITTI to quantify cross-dataset stability.
  - **Analyze class-wise degradation:** Identify whether vehicles, pedestrians, or cyclists suffer the largest performance drop.
  - Frame this milestone as the core novelty of the paper.
- **Deliverables:**
  - KITTI vs Waymo comparison table.
  - Generalization Ratio table.
  - Class-wise degradation chart.
  - Domain-shift analysis subsection.
  - External validation discussion draft.

Milestone 7: Robustness, Failure-Case & Safety-Oriented Analysis

- **Objective:**
  - **Analyze difficult perception cases:** Evaluate small objects, occlusion, crowded scenes, distant objects, and poor visibility where available.
  - **Prioritize safety-relevant errors:** Focus on false negatives for pedestrians and cyclists, not only average mAP.

- **Create failure-case galleries:** Show missed detections, false positives, localization errors, and class confusion.
- **Compare deployment trade-offs:** Relate accuracy to latency, FPS, and model complexity.
- Convert numerical results into practical recommendations for intelligent-vehicle perception.

- **Deliverables:**
  - Small/medium/large object analysis.
  - Occlusion and crowded-scene analysis.
  - Failure-case figure panel.
  - Safety-oriented error discussion.
  - Deployment suitability comparison.

Milestone 8: IEEE T-IV Paper Writing, Reproducibility & Submission Preparation

- **Objective:**
  - **Write the manuscript in IEEE style:** Structure the paper around motivation, related work, datasets, methodology, experiments, results, discussion, threats to validity, and conclusion.
  - **Strengthen the contribution:** Emphasize cross-dataset validation, representative detector families, deployment efficiency, and safety-relevant failures.
  - **Prepare reproducibility materials:** Clean code, configs, conversion scripts, evaluation scripts, and result tables.
  - **Check T-IV readiness:** Ensure the paper clearly connects to intelligent vehicles, automated driving, and real-time perception challenges.
  - Prepare final revision cycles before submission.

- **Deliverables:**
  - Complete IEEE-format manuscript.
  - Final tables and figures.
  - Reproducibility appendix.
  - Clean GitHub repository or supplementary material package.
  - Final submission checklist for IEEE T-IV.

Proposed Paper Structure

- **Abstract:** Problem, datasets, four models, cross-dataset validation, and key findings.
- **Introduction:** Autonomous-driving perception motivation, deployment constraints, and research gap.
- **Related Work:** Object detection, autonomous-driving datasets, and domain-shift evaluation.
- **Datasets:** KITTI preparation and Waymo subset construction.
- **Methodology:** Model setup, preprocessing, class mapping, and fairness rules.
- **Experimental Setup:** Hardware, metrics, hyperparameters, and reproducibility details.
- **Results:** KITTI in-domain results and Waymo external validation.
- **Discussion:** Generalization, safety-relevant failures, and deployment trade-offs.
- **Threats to Validity:** Dataset limitations, subset sampling, and implementation constraints.
- **Conclusion:** Main findings and future work.

Reference Sources to Use in the Paper

- IEEE Transactions on Intelligent Vehicles official scope page.
- KITTI Vision Benchmark Suite official object detection benchmark page.
- Waymo Open Dataset official documentation and perception dataset page.
- RetinaNet paper: Focal Loss for Dense Object Detection.
- RT-DETR paper: Real-Time Detection Transformer.
- Detectron2 official documentation/model zoo for Faster R-CNN and RetinaNet implementations.
- YOLO official documentation or implementation documentation for the selected YOLO version.